

Optimizing Industrial Heating with Parabolic Trough Collectors in Textile Manufacturing

Course Project Report for EN630

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Abstract

This study explores the potential of integrating and operating Solar Thermal Systems in Textile Industries. Concentrating Solar Collectors are selected considering the heating requirements in textile industrial application. We consider integration of solar heat collector instead of boiler for stenter machine and analyse the required parameters for the corresponding system. This analysis will cover the technical aspects of the system regarding heat and temperature requirements and give us a primitive inference about theoretical feasibility and aims to conclude the savings in energy and reduction in emissions. This analysis seeks to lay the foundation for solar technologies in reducing carbon footprint while simultaneously reducing non-renewable energy utilisation.

Keywords: Parabolic trough collector, Textile Industry, Solar thermal technology

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Chapter 1

Introduction

The textile industry in India plays a crucial role in the economy, significantly contributing to both exports and domestic markets. Valued at over USD 100 billion, the Indian Textile and Apparel Industry employs over 45 million people and accounts for 14% of the country's industrial production [1]. The sector also contributes approximately 15% to India's export earnings, with textiles, handicrafts, and apparel comprising 10.6% of India's total exports in FY22 [1]. For yarn production, the temperature requirement ranges from 85 to 95 °C, while fabric processing requires temperatures between 200 and 250 °C [3]. For industrial process heating applications, thus, the employment of concentrating solar collectors for Solar Industrial Process Heating (SIPH) is recommendable. Additionally, fabric processing requires steam at a temperature of 180 °C, and to ensure a consistent steam supply, the system design must include a steam accumulator. This integration not only enhances operational efficiency but also supports sustainable practices by optimizing energy usage. Implementing advanced technologies in the textile sector can further improve productivity, reduce costs, and strengthen India's competitive edge in the global market.

The study is based on a textile manufacturing facility which produces various types of fabric like cotton and polyester. Textile production starts with spinning process and continues with weaving, knitting, sizing, dyeing, and ends up with finishing. Among them sizing, dyeing and drying are wet processes. The dyeing and drying are the two processes that need large amounts of energy because of excessive heat they require to process the fabric. The dyeing process causes high amount water into the fabric. On the other hand, the drying process is used to take out the water from the fabric. Stenter machine is one of the most used machines to dry wet and humid fabric in textile process [4].

The objective of this report is to develop a design framework for integrating Parabolic Trough Collector (PTC) solar thermal technology within textile manufacturing facilities, addressing both radiations capture optimization and plant module implementation to enhance energy efficiency and reduce fossil fuel dependency in textile processing operations. PTCs efficiently deliver heat at 150–250 °C, matching the requirements of textile processes like yarn drying (80–120 °C) and steam generation (>150 °C).

The design integrates a PTC Solar Thermal System along with the Stenter machine to reduce boilers usage. The details of the requirement of both the Stenter Machine and Solar Thermal Plant are covered as following

1.1 Aim of the Project

To incorporate PTC into the existing textile industry:

- **Fuel Savings:**
To analyze the fuel savings on the incorporation of PTC into the existing system.
- **Emissions Reductions:**
To come up with reduction of emissions after incorporation of PTC in the industrial system.
- **Payback Period:**
Calculating the payback period upon incorporation of PTC system.
- **Application and Practical Relevance:**
To evaluate the economic and ecological benefits on incorporation of PTC by before and after comparison.

This paper will try to illustrate how the PTC system can be incorporated into the existing textile industry. Further it will try to delve into the techno-economic-ecological analysis of the new PTC system.

Chapter 2

Plant configuration

The Stenter machine along with the primary function of drying wet fabric, performs various other process used for stretching, and setting fabric. The flow diagram for operation parameters is depicted in Figure 2.1.

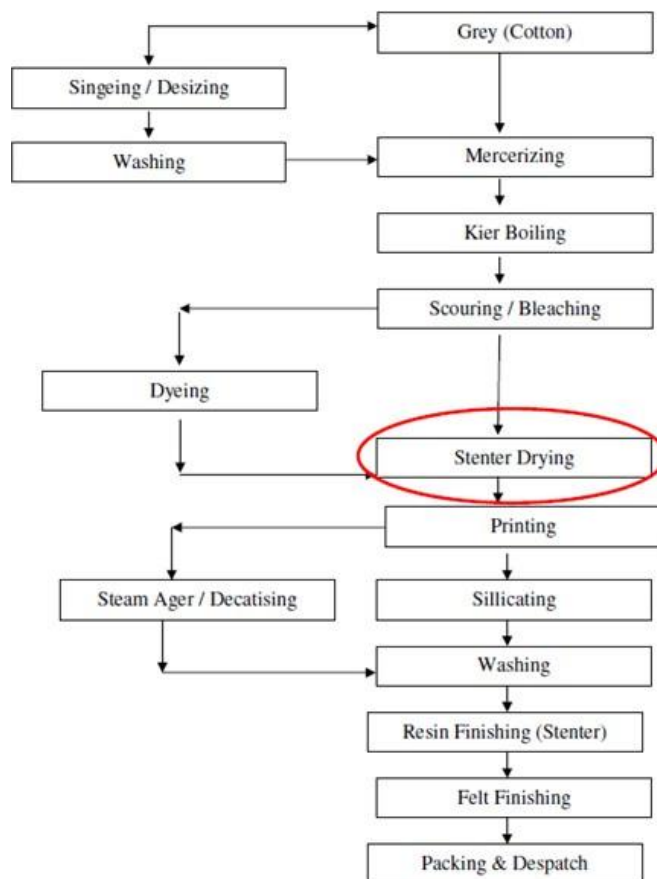


Figure 2.1: Textile Industry process flow chart [14]

Dry Chamber and Cooling roller are the two processes which require heating and temperature requirements, with GSM and type of cloth as design parameters. Considering the heating and temperature range requirement of 200 °C to 250 °C in the Stenter Machine, Parabolic Trough Collector is considered to provide the heat instead of boiler.

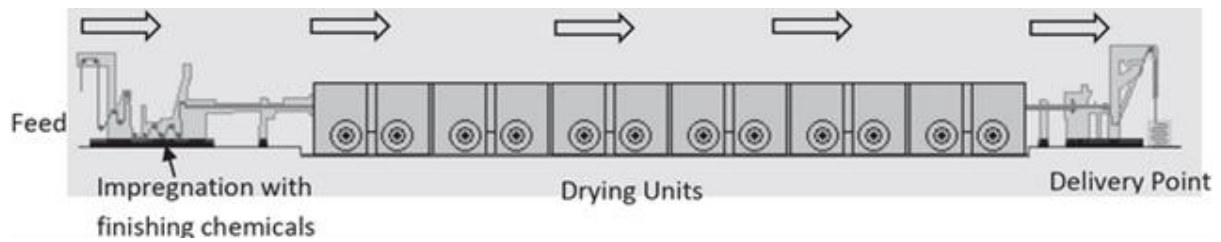


Figure 2.2: Schematic diagram of stenter machine [10]

The heating element requires 240 °C temperature in each of the 8 chambers which results in cloth being dried at a temperature of 225 °C as shown in Figure 2.3.



Figure 2.3: Stenter machine details

For the selected industrial operation, the specification and requirements of the complete production process are given in Table 2.1.

Assumptions:

- Mass flow rate of the fuel is constant throughout the day.
- CV of fuel: 18 MJ kg⁻¹ [13].
- Efficiency of the boiler: 90%.
- $\dot{m} = 2.8$ tonnes per day = 0.0648 kg s⁻¹.
- Total heat supplied by fuel in a day = $18 \times 2.8 \times 1000 = 50\,400$ MJ.
- Heat rate supplied by the boiler to the heat transfer fluid including efficiency = 933.33 kJ s⁻¹.

Table 2.1: Industrial plant details

S. No.	Parameter	Value
1	Timing	9:00 to 21:00 Hrs
2	Location	11°02' N; 77°18' E, Tirupur, Tamil Nadu - 641605
3	Product Production Rate	5.5 – 6 tonnes per day
4	Fuel	Wood
5	Fuel Consumption Rate	2.5 – 3 tonnes per day
6	Fuel Price	3400 / ton
7	Heat Transfer System	Oil
8	Stenter Required Temp	240 °C
9	Number of Chambers in the stenter	8

- Desired output from the solar field = 933.33 kJ s^{-1} .

Chapter 3

Modelling and Simulation

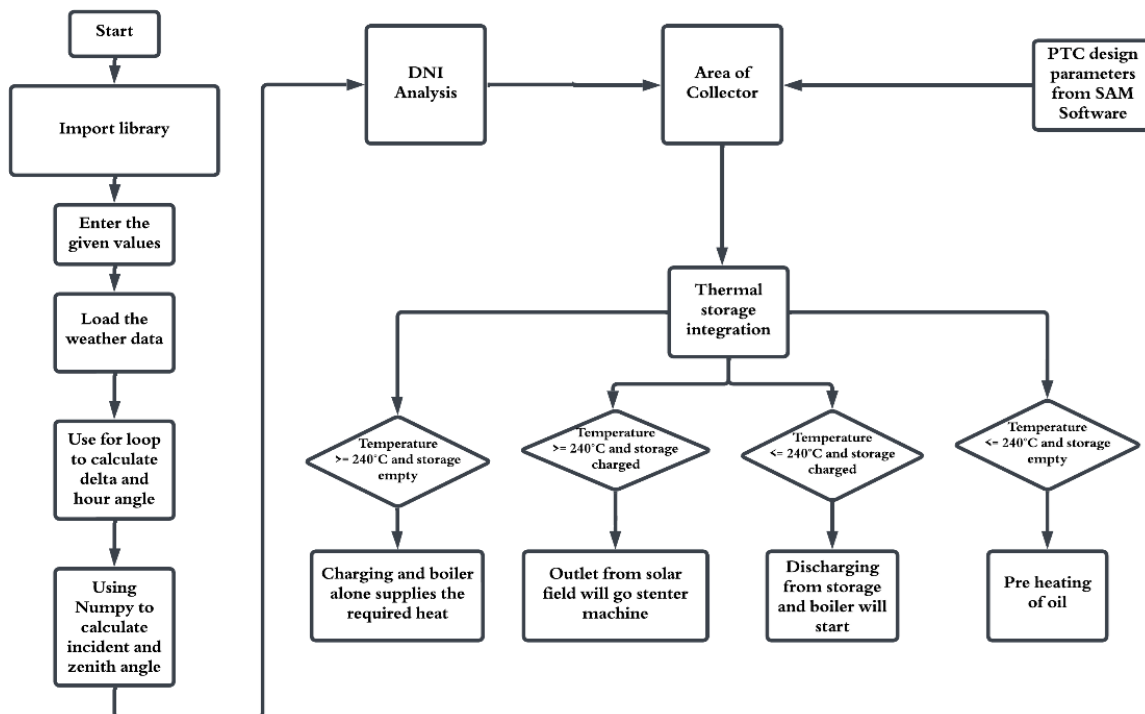


Figure 3.1: Flow chart for simulation

1. HT fluid having temperature more than 240 °C goes through heat exchanger before going to stenter machine.
2. Inlet stream of heat transfer fluid into the stenter machine and temperature is 240 °C.
3. HT fluid from stenter machine.
4. HT fluid from stenter and assumed temperature of HT fluid is 40 °C.
5. HT fluid pumping into the solar field.
6. HT fluid pumping into the solar field from oil drum.

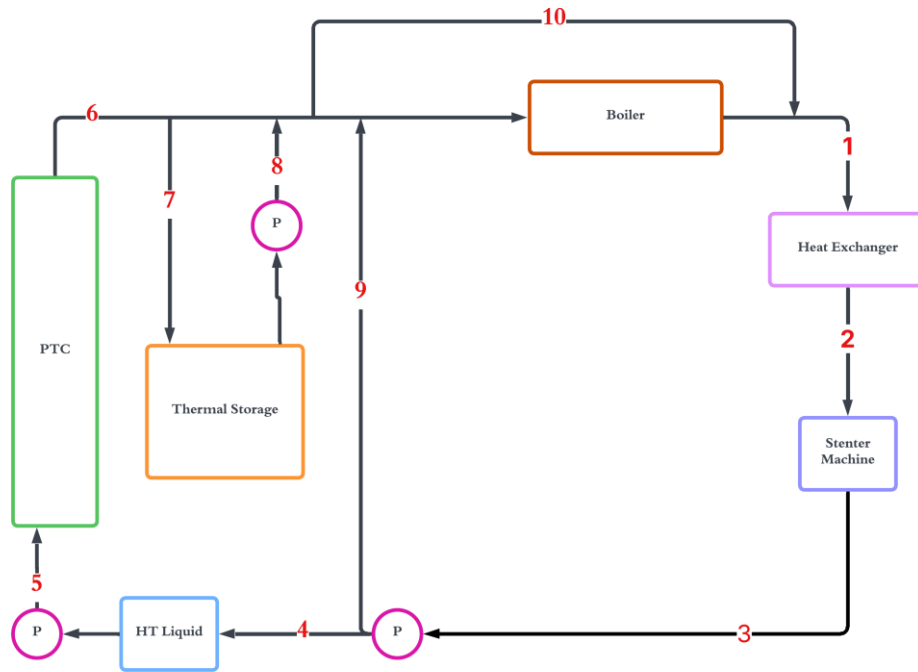


Figure 3.2: Schematic for integrated PTC in SIPH

7. HT fluid from solar field.
8. HT fluid into storage (Charging the storage).
9. HT fluid from storage (Discharging the storage).
10. During charging and non-sunshine hours of solar plant.
11. If outlet temperature exceeds desired temperature, HT fluid will not pass through boiler.

Chapter 4

Plant Model

The required aperture area for the parabolic trough solar field (A_{col}) was calculated as:

$$A_{\text{col}} = \frac{\dot{Q}_{\text{field}}}{\text{DNI} \cdot \eta_{\text{field}} \cdot \text{IAM} - U_{\text{loss}} \cdot (T_{\text{hft}} - T_{\text{amb}})} \quad (4.1)$$

where:

- $\dot{Q}_{\text{field}}$ is the required heat output from the solar field,
- DNI is the solar irradiance,
- IAM is the angle of incidence modifier,
- U_{loss} is the heat loss coefficient,
- ϑ is the incidence angle,
- T_{hft} is the average of solar field heat transfer fluid temperature,
- T_{amb} is the ambient temperature.

Table 4.1: Parabolic trough solar field design assumptions

Parameter	Symbol	Value
Design point	—	21st June at noon
Collector model	—	Luz LS-2
Receiver tube model	—	Schott PTR70 2008
Collector length	L_{col}	50 m
Collector width	W_{col}	5 m
Collector focal length	F_{col}	1.8 m
Collector spacing	$L_{\text{col,spa}}$	8 m
Solar field heat loss coefficient	U_{loss}	$0.1 \text{ W m}^{-2} \text{ K}^{-1}$
Solar field heat transfer fluid	—	Therminol VP-1
Solar field fluid outlet temperature	$T_{\text{hef,l}}$	240°C

From the design assumptions mentioned in Table 4.1, the angle of incidence (ϑ) is calculated using the following equation having relation between angle of declination (δ), zenith angle

(ϑ_z), and hour angle (ω):

$$\cos \vartheta = \sin \delta \sin \phi + \cos \delta \cos \phi \cos \omega \quad (4.2)$$

The incidence angle modifier (IAM) for parabolic trough collector model was calculated by:

$$\text{IAM} = \cos \vartheta \cdot 1 + \frac{\sin \vartheta}{360} \quad (4.3)$$

The parabolic trough solar field efficiency (η_{field}) was estimated as:

$$\eta_{\text{field}} = \eta_{\text{opt}} \cdot \eta_{\text{rec}} \cdot \eta_{\text{shd}} \cdot f_{\text{end}} \quad (4.4)$$

where:

- η_{opt} is the optical efficiency for the collector,
- η_{rec} is the receiver efficiency,
- η_{shd} is the shadowing efficiency,
- f_{end} is the end loss factor.

They are calculated using the following equations:

$$\eta_{\text{opt}} = f_{\text{trk}} \cdot f_{\text{geo}} \cdot \rho_{\text{col}} \cdot f_{\text{cln}} \quad (4.5)$$

$$\eta_{\text{rec}} = f_{\text{dst}} \cdot f_{\text{bel}} \cdot f_{\text{gl}} \cdot f_{\text{abs}} \quad (4.6)$$

$$\eta_{\text{shd}} = 1 - \frac{W_{\text{col}} \cdot \sin \vartheta}{L_{\text{col,spa}} \cdot \cos \vartheta_z} \quad (4.7)$$

$$f_{\text{end}} = 1 - \frac{F_{\text{col}} \cdot \tan \vartheta}{L_{\text{col}}} \quad (4.8)$$

Table 4.2: Parameters used in calculating the solar field efficiency

Parameter	Symbol	Value
Collector		
Tracking and twisting error factor	f_{trk}	0.99
Geometric accuracy of the collector mirror	f_{geo}	0.98
Mirror reflectivity	ρ_{col}	0.935
Average mirror cleanliness	f_{cln}	0.97
Receiver		
Dust losses factor	f_{dst}	0.98
Bellows shadowing factor	f_{bel}	0.963
Glass envelope transmissivity factor	f_{gl}	0.963
Absorber absorption factor	f_{abs}	0.96

Fuel requirement after integrating PTC:

$$m_{\text{fu}} = \frac{\dot{m}_{\text{flu}} \cdot C_p \cdot (T_{\text{htf},1} - T_{\text{o,PTC}})}{C_p} \quad (4.9)$$

where:

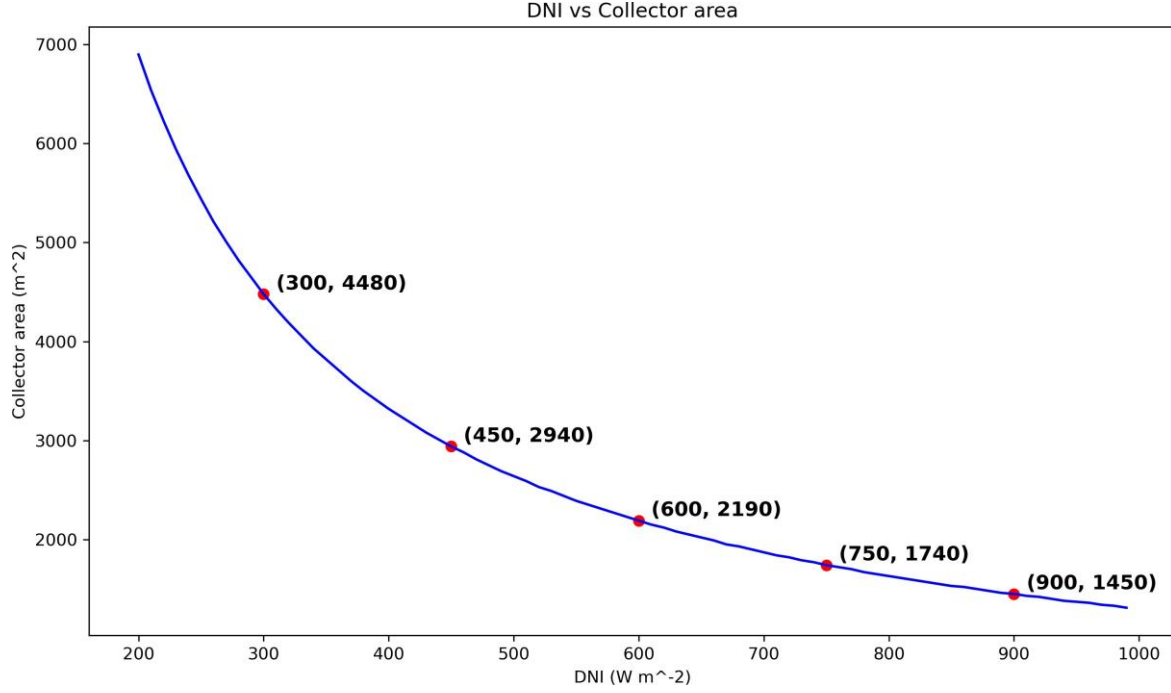


Figure 4.1: DNI versus collector area

- $\dot{m}_{fu}$ is the mass flow rate of fuel (kg s^{-1}),
- $\dot{m}_{flu}$ is the mass flow rate of thermal fluid (kg s^{-1}),
- $T_{htf,1}$ is the average temperature of the heat transfer fluid ($^{\circ}\text{C}$),
- $T_{o,PTC}$ is the outlet temperature of the PTC ($^{\circ}\text{C}$),
- C_p is the specific heat of fuel.

Table 4.3: Additional parameters

Parameters	Values
η_{opt}	0.975
η_{rec}	1.002
η_{shd}	0.8724
f_{end}	0.99
$f_{trk} \cdot f_{geo}$	0.9918
η_{field}	0.893
A_{col}	4480 m ²

Chapter 5

Thermal Storage Integration

Thermal energy storage (TES) is a common technology for heat storage, especially in concentrated solar power (CSP) facilities. In this thermal storage, instead of using molten salt, Therminol VP-1 is directly stored because it is directly used in the system as heat transfer liquid. Mass flow rate of heat transfer liquid is 4.51 kg s^{-1} , total mass required for 1-hour storage is 16 236 kg, density of Therminol VP-1 is 990.6 kg m^{-3} [11], thermal storage volume required for 1-hour storage of Therminol VP-1 is 16.39 m^3 .

Chapter 6

Optimization of collector area based on fuel savings

From the graph, collector area and wood savings have a positive correlation and a non-linear relationship. At the beginning, when collector area increases, wood savings increase steeply. For example, when the collector area increases from about 1870 m² to 2640 m², wood savings increase from approximately 398 tons / year to 462 tons / year, demonstrating a dramatic return on investment in this range. But beyond a certain point, about 4500 m², the marginal rate of increase in wood savings starts to fall, reflecting decreasing marginal returns. For example, expanding the area from 4480 m² to 6900 m² saves less wood from 524 tons/year to 565 tons/year. This implies that although increasing collector area still saves wood, the initial investment for the PTC will be so high, therefore, it is best to select the optimum collector area for the wood savings. Based on optimization findings, collector area of 4480 m² is taken into the account.

- Total aperture reflective area: 4480 m²
- Number of SCA in a row: 5
- Row spacing: 10 m
- Distance between SCA in a row: 1 m
- Total number of rows: 4
- SCA length: 50 m
- SCA width: 5 m
- Reflective area of one SCA: 235 m²
- Approximate number of SCAs required: 19
- Total solar field area: 14535 m²

6.1 Economic Analysis

The collector cost based on Indian market is | 15000 [12]. For 4480 m²; 6.72 crores of capital investment is required. The annual savings based on the area comes out to be

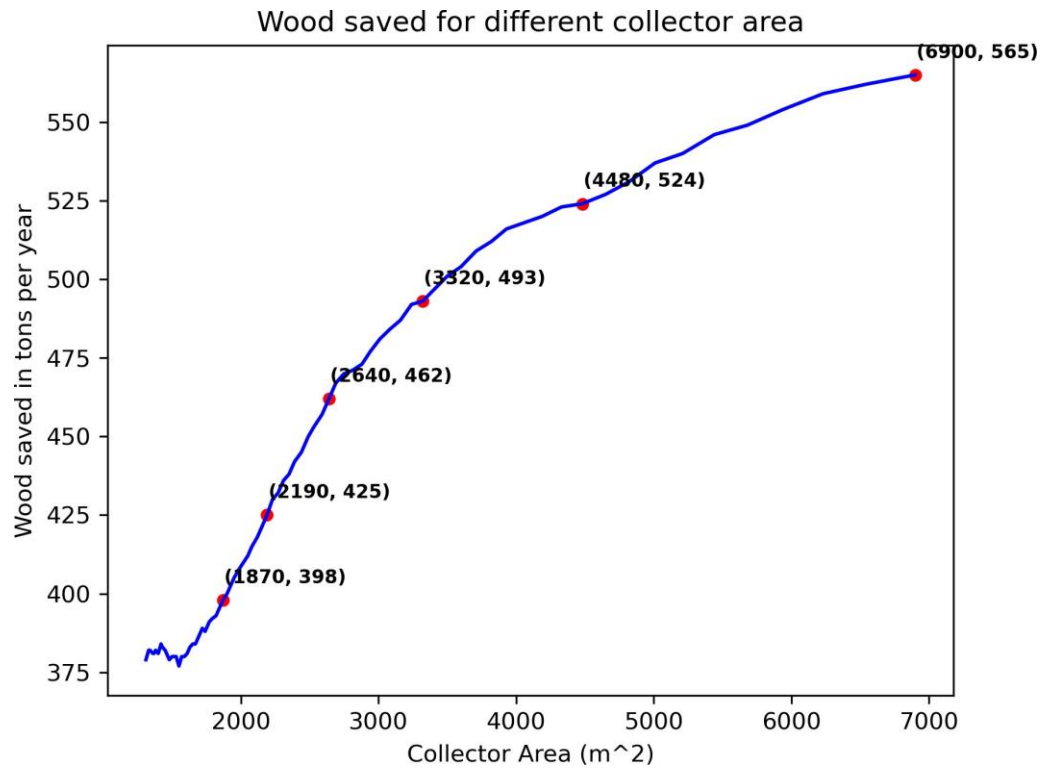


Figure 6.1: Wood savings for different collector area

17.82 Lakhs. Hence the Simple Payback Period based on Economic analysis is 38 years (without considering Therminol VP-1 cost).

Chapter 7

Results and Discussion

The integration of the Parabolic Trough Collector (PTC) system to the wood-type boiler saves 524 tonnes of wood per year (498 tonnes of wood being used as compared to 1022 before the integration).

A total of 4480 m² of PTC panels can be installed after the integration, enabling efficient capture of solar energy and its utilization.

To calculate the CO₂ emission reduction by using solar thermal energy instead of burning wood: Burning 1 tonne of wood emits approximately 1.75 tonne of CO₂ [8]. Therefore, burning 524 tonnes of wood would emit about $524 \times 1.75 = 917$ tonnes of CO₂. By using solar thermal instead of burning wood, the CO₂ emission reduction would be approximately 917 tonnes of CO₂ per year.

Chapter 8

Conclusion

The integration of the PTC into the wood-type boiler system has been responsible for energy savings and it is also contributing for both economic and environmental goodness. The cost saved demonstrate the effective role that solar energy can play in reducing operational costs and improving sustainability. Although the initial investment in PTC technology might be high, the long-term savings in fuel costs and the environmental benefits make it a worthwhile solution for reducing dependence on traditional fossil fuels or biomass resources.

Nomenclature

- **Abbreviations**

- PTC: Parabolic Trough Collector
- SIPH: Solar Industrial Process Heating
- TES: Thermal Energy Storage
- CSP: Concentrated Solar Power
- SIPH: Solar Industrial Process Heating
- SCA: Solar Collector Area

- **Symbols**

- A : Area, m^2
- T : Temperature, $^{\circ}\text{C}$
- U : Overall heat transfer coefficient, $\text{W m}^{-2} \text{K}^{-1}$
- IAM: Incidence Angle Modifier, degree
- c_p : Therminol VP-1 specific heat, $\text{kJ kg}^{-1} \text{K}^{-1}$

- **Greek symbols**

- δ : Delta
- ω : Hour angle

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